

KISANKAVACH

Karnataka

Master Database Design

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Farmer Credit. Benefits. Protection.

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Note: Page references are represented by section numbering to preserve stability across approved revisions.

1. Purpose and Scope

Defines the Karnataka pilot baseline for farmer-centred Kisan Credit Card enablement, benefit visibility, consent-driven data exchange, grievance handling, fraud protection and programme oversight.

In scope: farmer mobile experience, assisted service, participating-bank workflow, Government command centre, authorised integrations and operational reporting.

2. Guiding Principles

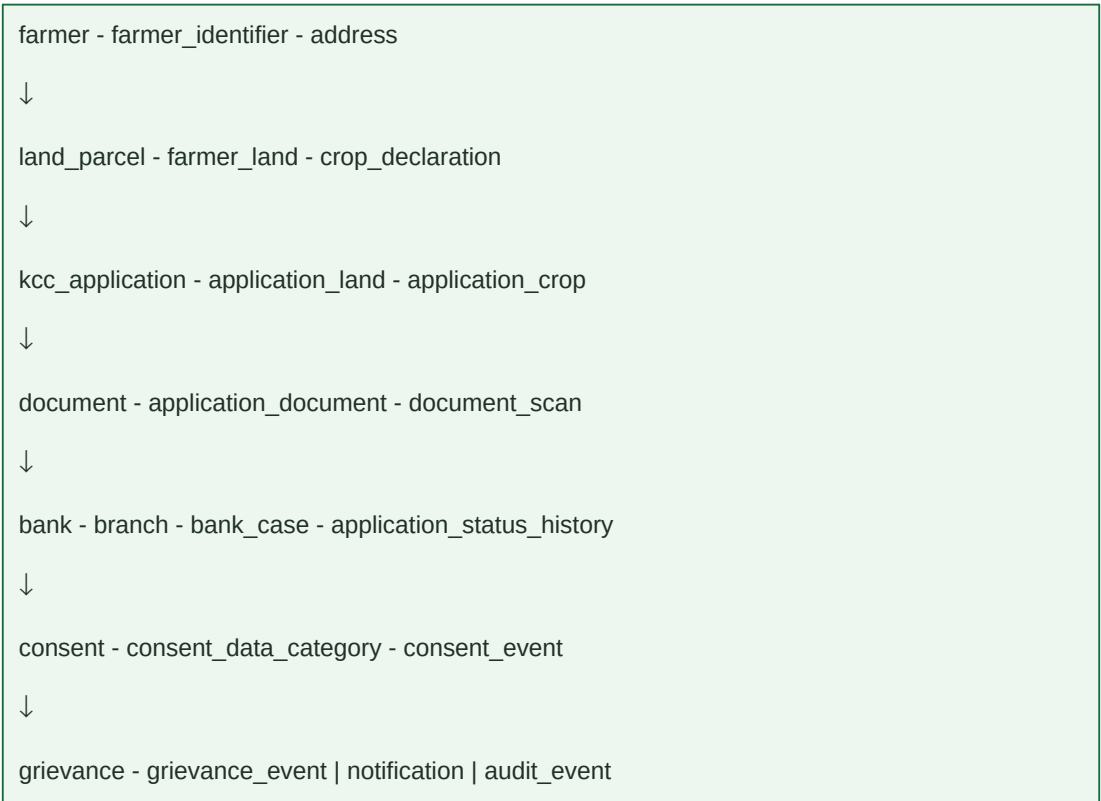
Kannada-first and low-literacy friendly; reuse verified information; explicit purpose-bound consent; least privilege; interoperability through adapters; configurable workflows and auditability.

The bank remains the final credit decision-maker. Government-originated and bank-originated data retain their source and provenance.

3. Data Design Principles

PostgreSQL is the system of record for KisanKavach operational state; external Government and bank facts retain provenance. Use UUID/ULID technical keys, separate human references, UTC timestamps, explicit status histories and immutable audit evidence.

4. Logical Domain Model



5. Core Tables

Table	Key fields	Purpose
farmer	farmer_id, display_name, mobile_hash, status	Platform farmer profile

Table	Key fields	Purpose
external_reference	source, reference_value, retrieved_at	Authoritative-system linkage
land_parcel	parcel_id, survey_no, extent, source	Land snapshot and provenance
crop_declaration	crop_id, season, area, source, status	Crop evidence
kcc_application	application_id, public_ref, bank_id, amount, status	KCC aggregate root
application_status_history	from_status, to_status, actor, occurred_at	Immutable lifecycle
document	document_id, object_key, checksum, scan_status	Secure file metadata
consent	purpose, recipient, granted_at, expires_at, status	Consent evidence
grievance	category, priority, owner, due_at, status	Case management
audit_event	actor, action, resource, correlation_id, event_at	Protected evidence

6. Application Relationships

A farmer may have multiple applications over time. Each application snapshots selected land/crop and verified values to preserve the submitted record. Status is materialised on the application for fast reads and fully represented in append-only history.

7. Status and Reference Data

Entity	Representative controlled values
Application	DRAFT, SUBMITTED, DOCUMENT_CHECK, BANK_REVIEW, SANCTIONED, REJECTED, DISBURSED, ACTIVE, CLOSED
Consent	REQUESTED, GRANTED, DENIED, EXPIRED, WITHDRAWN
Document scan	QUARANTINED, SCANNING, CLEAN, REJECTED
Grievance	OPEN, ASSIGNED, IN_PROGRESS, RESOLVED, CLOSED, REOPENED

8. Indexes, Constraints and Concurrency

Unique public application reference; composite indexes on bank/branch/status/updated_at and farmer/status; foreign keys for ownership; check constraints for positive amount/extent; optimistic version column on mutable aggregates; partial unique rules for active applications where policy requires.

9. Security and Privacy

Separate sensitive identifiers from display data; encrypt high-risk columns where needed; use row/scope enforcement in services; restrict raw production access; log reads and exports; tokenize or hash search identifiers; never store secrets in tables.

10. Retention, Partitioning and Archival

Partition high-volume audit, notification and status-event tables by time. Retention is category- and purpose-specific. Expiry jobs create auditable disposition records; legal or regulatory holds override deletion.

11. Migration and Quality

Schema changes are versioned, backward compatible during rolling deployment, reviewed and tested on production-like volumes. Data-quality checks cover duplicates, orphan references, invalid transitions, missing provenance and consent/application inconsistencies.

12. Backup and Recovery

Encrypted automated backups and point-in-time recovery are required. Restore drills validate integrity, application compatibility and documented RPO/RTO. Backup access and deletion are tightly restricted and audited.

End of controlled baseline